

- 7 A particle P is projected from a point O with speed U at an angle 45° above the horizontal and moves freely under gravity.

(a) State the vertical and horizontal components of velocity at time t . [1]

[illegible]

At time T , particle P is moving at an angle of 60° below the horizontal.

(b) Show that $T = \frac{U}{2g}(\sqrt{2} + \sqrt{6})$. [3]

[illegible]



At time T , the particle strikes a smooth horizontal plane at a point which is a horizontal distance D from O and a vertical distance H below O .

(c) Find the ratio $H : D$. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

After striking the horizontal plane, P rebounds with speed w . The coefficient of restitution between P and the plane is $\frac{2}{3}$.

(d) Find w in terms of U . [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

